

StudyHub

Project Documentation Report

A MERN-based Educational Platform
for Notes Sharing and Student Collaboration

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1. Introduction

StudyHub is a MERN stack web application for educational content sharing and collaboration.

1.1 Existing System and Need for System

Traditional systems lack:

- Centralized note sharing platforms
- Real-time collaboration features
- Community-driven learning

1.2 Scope of System

- User authentication and authorization
- Notes upload/download system
- Community forums with real-time chat
- Poll and notification systems

1.3 Operating Environment

Hardware:

- Processor: Intel Core i3+
- RAM: 4GB minimum
- Storage: 500MB+

Software:

- Node.js v14+
- MongoDB v4.4+
- Modern web browser

2. Proposed System

2.1 Feasibility Study

Technical: MERN stack provides robust framework

Economic: Open-source technologies reduce costs

Operational: Intuitive UI ensures easy adoption

2.2 Objectives

- Centralized educational content platform
- Real-time student-teacher collaboration
- Organized subject-specific communities
- Secure authentication system

2.3 Users of System

- Students: Upload/download notes, join communities
- Teachers: Share content, manage materials

3. Analysis and Design

3.1 System Requirements

Functional:

- User registration and authentication
- Notes upload with metadata
- Search and filter functionality
- Community forums and messaging
- Poll creation and voting

Non-Functional:

- Performance: <3s page load
- Scalability: 10,000+ users
- Security: Encrypted data
- 99.5% uptime

3.2 Module Hierarchy

Frontend:

- Authentication, Home, Notes, Communities, Profile

Backend:

- Auth Service, Notes Management, Community Service
- Messaging, Notifications, Polls

3.3 Sample Input and Output Screens

Figure 1: Home Page - Landing page with features

[Screenshot: Home Page - Landing page with features]

Figure 2: Login Page - User authentication

[Screenshot: Login Page - User authentication]

Figure 3: Notes Page - Browse and search notes

[Screenshot: Notes Page - Browse and search notes]

Figure 4: Upload Page - Upload study materials

[Screenshot: Upload Page - Upload study materials]

Figure 5: Communities - Available forums

[Screenshot: Communities - Available forums]

Figure 6: Community Detail - Real-time chat

[Screenshot: Community Detail - Real-time chat]

Figure 7: User Profile - Statistics and content

[Screenshot: User Profile - Statistics and content]

Figure 8: Teacher Dashboard - Content management

[Screenshot: Teacher Dashboard - Content management]

4. Coding

4.1 Technology Stack

- Frontend: React 19, TailwindCSS, Material-UI
- Backend: Node.js, Express 5, MongoDB
- Real-time: Socket.IO 4.8
- Auth: JWT, Bcrypt
- Storage: Cloudinary

Key Features:

```
// Server Setup
import express from 'express';
import mongoose from 'mongoose';
const app = express();
app.use(compression());

// Routes
app.use('/api/auth', authRoutes);
app.use('/api/notes', notesRoutes);
```

5. Limitations

- Limited to PDFs (no video support)
- Requires stable internet
- 50MB file size limit
- Basic search functionality
- No offline access

6. Proposed Enhancements

- Video lecture support
- AI-powered recommendations
- Mobile applications
- Offline PWA mode
- Advanced analytics
- Gamification features
- Multi-language support

7. Conclusion

StudyHub successfully provides a comprehensive educational platform with:

- Seamless user experience
- Real-time collaboration
- Scalable architecture
- Secure content management

The platform demonstrates strong potential for growth and has successfully met its core objectives.

Deployed at: <https://studyhub4all.netlify.app>

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